$$x = \frac{1}{t^2-1},\ x+1 = \frac{t^2}{t^2-1},\ t = \frac{\sqrt{x+1}}{\sqrt{x}}$$

$$\sqrt{x(x+1)} = \frac{t}{t^2-1}\ dx = -\frac{2t}{(t^2-1)^2}\,dt$$

$$\begin{aligned}\int \frac{1}{\sqrt{x(x+1)}}\,dx &= \int \frac{t^2-1}{t}\cdot\left(-\frac{2t}{(t^2-1)^2}\right)\,dt \\ &= -2\int \frac{1}{t^2-1}\,dt \\ &= \ln\left|\frac{t+1}{t-1}\right| + c\end{aligned}$$

$$\frac{t+1}{t-1} = \frac{\sqrt{x+1}+\sqrt{x}}{\sqrt{x+1}-\sqrt{x}} = \left(\sqrt{x+1}+\sqrt{x}\right)^2$$

$$\int \frac{1}{\sqrt{x(x+1)}}\,dx = 2\ln\left(\sqrt{x}+\sqrt{x+1}\right) + c$$

$$y = \frac{1}{\sqrt{x}} - \frac{1}{\sqrt{x+1}}$$

$$\begin{aligned}V &= \pi\int_{1/8}^{9/16} y^2\,dx \\ &= \pi\int_{1/8}^{9/16}\left(\frac{1}{x}+\frac{1}{x+1}-\frac{2}{\sqrt{x(x+1)}}\right)\,dx \\ &= \pi\left[\ln x + \ln(x+1) - 4\ln\left(\sqrt{x}+\sqrt{x+1}\right)\right]_{1/8}^{9/16} \\ &= \pi\left(\ln\frac{225}{256} - 4\ln 2 - \left(\ln\frac{9}{64} - 4\ln\sqrt{2}\right)\right) \\ &= \pi\left(\ln\frac{225}{4096} - \ln\frac{9}{256}\right) \\ &= 2\pi\ln\frac{5}{4}\end{aligned}$$

This solution was submitted by **Jon**

Published and refined by **Jon**

Last updated: June 25, 2026